

**CAMBRIDGE INFOTECH**

CORPORATE TRAINING SERIES

# **Prompt Engineering**

Comprehensive Assignment: Prompt Generation Practice — Completed Answers

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*Cambridge Infotech | Prompt Engineering Training Program | Assignment Booklet*

## Module 1 · Introduction to Generative AI

*AI, ML, Deep Learning, LLMs, Tokens & Why Prompting Matters*

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### Q1. Prompt

Act as a computer science professor teaching first-year college students who have no prior background in AI.

Explain Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI as four layered ideas, where each concept is a more specific subset of the one before it.

For each of the four concepts, give a simple one-line definition followed by one real-world example that a student would recognize (for instance, a voice assistant, a spam filter, a face-unlock feature, or an AI image generator).

Use plain, everyday language, avoid heavy technical jargon, and keep the explanation of each concept to no more than three to four sentences.

Present the four concepts as a short nested list so the 'layers' relationship is visually clear.

### Q2. Prompt

Act as an AI teacher explaining to a complete beginner with no technical background how an AI model processes a message they type.

Describe the three stages the model goes through: tokenization, understanding context, and predicting the next tokens.

Use a single simple analogy that runs through all three stages consistently, for example comparing the AI to someone reading a sentence word by word, understanding the story so far, and guessing what comes next.

Keep technical terms to a minimum, and when a technical term is used, immediately explain it in plain words.

Limit the explanation to about 150 words and end with one sentence summarizing why this process matters for getting good responses.

### Q3. Prompt

Act as an AI researcher explaining to a curious non-technical reader why a large language model does not simply search the internet and copy an answer.

Explain clearly that the model generates a response word by word (token by token) based on patterns it learned during training, rather than retrieving stored text.

Describe the role of probability in this process: at each step the model calculates the likelihood of many possible next words and selects one, rather than 'looking up' a fixed answer.

Include a short example showing how the same question could produce slightly different but still correct phrasings, to illustrate this probabilistic behavior.

Keep the explanation to around 200 words and avoid mathematical notation.

### Q4. Prompt

Act as an AI trainer explaining the concept of a 'context window' to someone who has just started using AI chat tools.

Explain what a context window is, why every AI model has a fixed limit on it, and what happens to the conversation once that limit is exceeded.

Include a simple numeric example, such as a model with an 8,000-token context window, and show what happens when the conversation history grows beyond that number (for example, the oldest messages being dropped or summarized).

Write the explanation in plain, non-technical English suitable for a general audience.

Keep the total response under 180 words and end with one practical tip for managing long conversations.

### Q5. Prompt

Act as an AI educator preparing a comparison handout for students learning how language models work.

Compare 'Training' and 'Inference' side by side using a two-column table format, with one column for Training and one for Inference.

For each phase, describe what actually happens (for example, learning patterns from massive datasets versus generating a response to a live prompt), when it occurs in the model's lifecycle, and how much computing power is typically involved.

Keep each table cell to one or two short sentences so the comparison stays easy to scan.

Add a one-sentence summary below the table stating the key difference in plain language.

### Q6. Poor vs Improved Prompt

Poor version: "What is Machine Learning?"

Improved version: "Act as a friendly technology teacher explaining Machine Learning to a group of high-school students who have never studied computer science. In under 150 words, explain what Machine Learning is, and include one real-world example, such as movie recommendation systems, that the students would recognize from daily life."

One-sentence explanation: The improved version produces a better response because it defines a clear role, a specific audience, a length constraint, and an explicit request for a relatable example, which together remove ambiguity about tone, depth, and content that the vague version leaves entirely open.

## Module 2 · Prompt Engineering Fundamentals

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*Anatomy of a Prompt: Role, Task, Context, Instructions, Constraints, Output Format*

### Q7. Prompt

Role: Act as a friendly technology consultant who specializes in explaining IT concepts to non-technical business owners.

Task: Explain what 'Cloud Computing' is and why it matters for a small or medium-sized business.

Context: The audience runs a retail business, has no IT background, and is currently deciding whether to move their store's data and software online instead of using physical servers.

Instructions: Cover the basic definition, two or three practical benefits (such as cost savings and remote access), and one common concern (such as data security), using simple analogies instead of technical terms.

Constraints: Keep the explanation under 200 words and avoid acronyms unless you immediately explain them. Output Format: Present the answer as a short introductory paragraph followed by three bullet points for the benefits.

### Q8. Rewritten Prompt

Original vague prompt: "Write something for my manager."

Rewritten prompt: Act as a professional employee drafting a short internal email to your direct manager.

The purpose of the message is to request two additional days of remote work next week due to a personal commitment, and to reassure the manager that all deadlines will still be met.

Write the email for a manager who prefers direct, no-nonsense communication, keep it under 120 words, and use a polite, professional, and confident tone throughout.

Output Format: Provide a subject line and a body with a clear opening request, one supporting reason, and a closing line offering to discuss further if needed.

### Q9. Prompt (Role only)

Act as an experienced Human Resources Manager with over ten years of experience drafting internal company policies.

Based on this role, draft a short policy note explaining the company's work-from-home guidelines, including eligibility, expected working hours, and communication expectations.

Write in a professional but approachable tone that reflects how a seasoned HR manager would communicate with employees.

Keep the note concise enough to fit on a single page.

Structure it with a short title, a brief introduction, and three to four clearly labeled guideline points.

### Q10. Prompt (Context)

Context: I am a final-year engineering student about to attend a campus placement interview for a software engineering role at a mid-sized technology company. I have completed two internships, built two personal coding projects, and I am nervous about interviews but eager to make a strong first impression.

Task: Using this context, help me craft a confident and concise answer to the common interview question 'Tell me about yourself.'

Instructions: Structure the answer to briefly cover my educational background, relevant project or internship experience, and what I am looking for in this role, in that order.

Constraints: Keep the spoken answer to about 45-60 seconds when read aloud (roughly 120-150 words), and avoid sounding rehearsed or overly formal.

Output Format: Provide the answer as a natural, first-person script I can practice saying out loud.

### Q11. Prompt (Constraints only)

Explain 'Blockchain Technology' in exactly 80 words.

Do not use any technical jargon such as 'hash', 'consensus', 'node', or 'cryptography' — describe the concept using only everyday language and simple comparisons.

The explanation should still convey the core idea of a shared, tamper-resistant digital record that many people can trust without needing a single central authority.

Write it as a single flowing paragraph rather than a list.

Double-check the final word count before responding and adjust if it is not exactly 80 words.

### Q12. Reusable Prompt Template

Act as a [ROLE] explaining [TASK] to [AUDIENCE].

Context: [CONTEXT — briefly describe the situation, background, or purpose behind why this explanation is needed].

Instructions: [INSTRUCTIONS — specify what points must be covered, the depth of detail required, and any examples or analogies that should be included].

Constraints: [CONSTRAINTS — specify word or length limits, tone requirements, and any words, jargon, or content to avoid].

Output Format: [FORMAT — specify whether the response should be a paragraph, bullet points, a table, a numbered list, or another structure].

## Module 2 (continued) · Basic Prompting Techniques

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*Zero-Shot, One-Shot, Few-Shot, Role, Context, Delimiters & Iteration*

### Q13. Zero-Shot Prompt

Classify the sentiment of the following customer review as Positive, Negative, or Neutral.

Review: "The delivery was fast, but the packaging was damaged."

Do not provide any explanation or reasoning — respond with only a single word representing the classification.

Base your judgment purely on the overall tone of the review, weighing both the positive and negative elements mentioned.

If the review expresses genuinely mixed sentiment, classify it as Neutral rather than forcing it into Positive or Negative.

### Q14. One-Shot Prompt

Classify customer feedback messages into either 'Complaint' or 'Compliment' based on the example below.

Example: Feedback: "I waited three weeks for a replacement part and no one responded to my emails." → Complaint

Now classify this new feedback using the same logic and format: "Your support team resolved my billing issue within minutes — really impressed!"

Respond with only the category label, matching the style of the example exactly.

If the feedback contains no clear complaint or compliment, state 'Unclear' instead of guessing.

### Q15. Few-Shot Prompt

Classify each sentence's tone as Formal or Informal based on the three labeled examples below.

Example 1: "Please find attached the requested documentation for your review." → Formal

Example 2: "Hey, can you send me that file when you get a sec?" → Informal

Example 3: "We regret to inform you that your application has not been successful at this time." → Formal

Now classify these two new sentences using the same pattern: (1) "lol that meeting ran way too long" (2) "Kindly confirm your availability for the scheduled call." Respond with only the sentence number and its label.

### Q16. Prompt with Delimiters

Summarize the text enclosed in triple quotes below into exactly three sentences, focusing only on the main argument and key conclusion.

Do not summarize or reference anything outside the triple-quoted block, and do not repeat the instructions in your answer.

""""[Insert the full block of sample text to be summarized here]""""

Return only the three-sentence summary as plain text, with no headings or bullet points.

One-sentence explanation: Delimiters reduce confusion for the model because they create a clear visual and structural boundary between the instructions and the content to be processed, preventing the model from misreading part of the instructions as part of the text or vice versa.

### Q17. Three-Step Prompt Refinement

(a) Poor prompt: "Explain databases."

(b) Improved prompt: "Act as a technology teacher and explain what a database is to a high-school student who has never studied computer science, using simple, everyday language and one relatable analogy such as a digital filing cabinet."

(c) Final prompt: "Act as a technology teacher and explain what a database is to a high-school student who has never studied computer science, using simple, everyday language and one relatable analogy. Limit the explanation to exactly 100 words, and structure the output as a short introductory sentence followed by three bullet points covering what a database stores, why it is useful, and one real-world example."

Each version adds one more layer of the prompt anatomy — audience and style in step (b), then length and output structure in step (c) — progressively narrowing the range of acceptable responses.

This progression demonstrates how iterative refinement steadily improves response quality and predictability.

### Q18. Role + Context Prompt

Act as an experienced Career Counsellor who has guided hundreds of students into successful careers across technology, business, and creative fields.

Context: The student is a third-year undergraduate majoring in Computer Science, has strong skills in Python and data analysis, enjoys solving logical puzzles, has completed a summer internship in a data analytics team, and is unsure whether to pursue a technical role, a research path, or a management-track career.

Task: Based on this background, recommend three suitable career paths for this student.

Instructions: For each recommendation, briefly explain why it fits the student's skills and interests, and mention one concrete next step (such as a certification, project, or skill) to pursue it.

Output Format: Present the three recommendations as a numbered list, each with a one-line title followed by two to three sentences of explanation.

## Module 4 · Advanced Prompt Engineering Techniques

*Chain-of-Thought, Self-Consistency, Step-Back, Decomposition, Chaining, Self-Reflection, ReAct*

### Q19. Chain-of-Thought Prompt

Solve the following problem step by step, showing each calculation clearly before giving the final answer.

Problem: A shop offers a 15% discount on a product priced at ₹3,500, and then adds 5% tax on the discounted price. What is the final amount to be paid?

First calculate the discount amount and the discounted price, then calculate the tax on that discounted price, and finally add the tax to get the total amount.

Show each step as a separate labeled line (for example, 'Step 1: Calculate discount amount') rather than jumping straight to the final number.

Clearly state the final answer on its own line at the end, rounded to two decimal places if needed.

### Q20. Self-Consistency Prompt

Solve the following problem using two different solution approaches, and then compare the two results.

Problem: A car travels 240 kilometers in 4 hours at a constant speed, then continues for another 90 kilometers at the same speed. How long does the second part of the journey take?

Approach 1: Calculate the speed first using the initial distance and time, then use that speed to find the time for the second distance.

Approach 2: Set up and solve the problem using a proportion (ratio) between distance and time directly, without explicitly calculating speed as an intermediate step.

After showing both approaches, compare their final answers, confirm whether they agree, and state the final, most reliable answer with a brief justification for why it can be trusted.

### Q21. Step-Back Prompt

First, step back and identify the general economic principle that explains why two products with very similar features can be priced very differently — think about factors such as brand perception, marketing, quality assurance, and customer experience beyond raw specifications.

State this general principle clearly in two to three sentences before moving to any specific example.

Then, apply that general principle to a specific example: explain why two smartphones with nearly identical hardware specifications can differ in price by 30% or more.

Make sure the specific example explanation directly refers back to the general principle you identified, rather than introducing unrelated reasoning.

Keep the entire response under 200 words, split into a 'General Principle' section and an 'Applied Example' section.

### Q22. Prompt Decomposition Sequence

Prompt 1: Act as a social media marketing strategist. Identify the target audience and the two most suitable social media platforms for a newly opened bakery in a mid-sized city, with a brief justification for each choice.

Prompt 2: Based on that target audience and platform choice, suggest four content pillars (recurring content themes) the bakery should post about, with one example post idea for each pillar.

Prompt 3: Create a two-week posting calendar for the bakery, specifying the day, platform, content pillar, and a short caption idea for each planned post.

Prompt 4: Suggest a simple engagement and growth strategy for the bakery's social media, including one low-budget paid promotion idea and one organic engagement tactic (such as a giveaway or collaboration).

Prompt 5 (optional synthesis): Combine the outputs from Prompts 1 to 4 into a single one-page social media marketing plan summary for the bakery owner.

### Q23. Prompt Chain (3 steps)

Prompt 1 (Extraction): "Read the following long article and extract the five to seven most important key points as a bulleted list, focusing only on facts, findings, and conclusions, and ignoring introductory or background text. Article: [insert article text]."

Prompt 2 (Summarization): "Using only the key points extracted below, write a concise 100-150 word summary of the article in flowing paragraph form, suitable for someone who has not read the original. Key points: [insert output from Prompt 1]."

Prompt 3 (Presentation Outline): "Using the summary below, create a 5-slide presentation outline, where each slide has a short title and two to three bullet points capturing the essential message for that slide. Summary: [insert output from Prompt 2]."

Each prompt explicitly instructs the model to use only the output of the previous step as its input, ensuring the chain stays consistent and grounded.

This chaining approach breaks a large task into focused sub-tasks, improving accuracy and making each intermediate output independently reviewable.

### Q24. Self-Reflection Prompt

First, answer the following question directly: "What are the main causes of climate change and what are two realistic solutions individuals can adopt?"

After providing your initial answer, review it critically for factual errors, missing important information, and any explanations that might be unclear to a general audience.

List the specific issues you found in your own answer under a heading called 'Self-Review Notes', even if the issues are minor.

Then provide an improved, corrected version of the answer under a heading called 'Improved Answer', incorporating the fixes identified in your review.

Keep both the initial answer and the improved answer under 150 words each so the improvement is easy to compare.

### **Q25. Critique-and-Revise Prompt**

First, draft a short apology email to a client explaining that their order has been delayed by five business days due to a supplier issue, and reassure them of the new expected delivery date.

After drafting the email, critique your own draft specifically for tone (is it appropriately apologetic without being overly submissive?) and clarity (is the delay reason and new timeline stated unambiguously?).

List at least two specific weaknesses found in the critique, even if they are minor phrasing issues.

Then produce a revised, final version of the email that addresses each weakness identified in the critique.

Present the response in three clearly labeled sections: 'Initial Draft', 'Critique', and 'Revised Version', with the email kept under 120 words in both versions.

## **Module 5 · Prompting for Different Tasks**

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*Generation, Summarization, Classification, Extraction, Translation, Rewriting, Q&A*

### **Q26. Text Generation Prompt**

Act as a professional content writer specializing in corporate LinkedIn posts.

Write a 300-word LinkedIn post announcing the launch of a new AI training program at Cambridge Infotech, aimed at working professionals who want to upskill in artificial intelligence and prompt engineering.

Highlight the program's practical, hands-on approach, mention that it suits both technical and non-technical professionals, and include a clear call-to-action encouraging readers to enroll or learn more.

Use an engaging, motivational, and professional tone appropriate for a corporate LinkedIn audience, and include 3-4 relevant hashtags at the end.

Structure the post with a strong opening hook, two to three short paragraphs, and a closing call-to-action line.

### **Q27. Summarization Prompt**

Condense the following long research article into exactly 5 bullet points.

Focus only on the key findings and conclusions of the research, and completely ignore background information, literature review, or methodology details unless they are essential to understanding a finding.

Each bullet point should be a single concise sentence, written in plain language suitable for a busy executive skimming the summary.

Do not add any interpretation, opinion, or information that is not explicitly present in the article.

Article: [insert full research article text here].

### **Q28. Classification Prompt**

Classify each of the following five customer support messages into exactly one of three categories: 'Complaint', 'Inquiry', or 'Feedback'.

Messages: (1) "My order arrived broken and I want a refund immediately." (2) "Do you offer international shipping to Canada?" (3) "Just wanted to say your customer service team was amazing today!" (4) "I've been charged twice for the same order, please fix this." (5) "What are your customer support hours on weekends?"

Return only the category for each message, in the format 'Message number: Category', with no additional explanation or reasoning.

If a message could fit more than one category, choose the single most dominant intent expressed in the message.

Present the final output as a simple numbered list matching the order of the messages given.

### **Q29. Information Extraction Prompt**

Extract the following four fields from the candidate profile below: Name, Job Title, Years of Experience, and Key Skills.

Candidate profile: "Ananya Rao is a Senior Data Analyst with 6 years of experience in the analytics industry. She specializes in Python, SQL, Power BI, and statistical modeling, and has led data-driven projects for retail and finance clients."

Return the extracted information as a clean, labeled list with one field per line, in the exact order requested.

For 'Key Skills', list all skills mentioned as a comma-separated list rather than a paragraph.

If any field is not explicitly mentioned in the profile, write 'Not specified' instead of guessing or inferring a value.

### **Q30. Translation Prompt**

Translate the following paragraph about renewable energy into Hindi.

Paragraph: "Renewable energy sources such as solar, wind, and hydropower are becoming increasingly important as the world moves away from fossil fuels. These technologies help reduce carbon emissions while providing sustainable power for homes and industries."

Preserve technical terms such as 'solar', 'wind', and 'hydropower' in a natural way that a Hindi-speaking general audience would still understand, rather than forcing overly literal or obscure translations.

Use natural, everyday conversational Hindi rather than overly formal or literary language, so the translation reads smoothly when spoken aloud.

Return only the translated paragraph, with no English text included in the final output.

### Q31. Rewriting/Transformation Prompt

Rewrite the following casual message in three different tones: Professional, Friendly, and Formal.

Original message: "hey can u pls send the report by tmrw eod, thanks"

For the Professional version, use clear, workplace-appropriate language without being overly stiff. For the Friendly version, keep it warm and casual but still clearly understandable. For the Formal version, use complete, polished sentences suitable for a formal business letter or official communication.

Expand all abbreviations (such as 'pls', 'tmrw', 'eod') into full words in every version.

Present the three rewrites as clearly labeled separate paragraphs, one under each tone heading.

### Q32. Context-Based Q&A Prompt

Background context: "The Great Barrier Reef, located off the coast of Queensland, Australia, is the world's largest coral reef system, stretching over 2,300 kilometers. It is home to thousands of species of marine life, including fish, sharks, and sea turtles. In recent decades, rising ocean temperatures have caused repeated coral bleaching events, threatening the reef's long-term survival."

Using only the background context provided above, answer the following question: "What is the primary threat to the Great Barrier Reef mentioned in the passage?"

Do not use any outside knowledge beyond what is stated in the context, even if you know additional facts about the topic.

If the answer cannot be fully determined from the given context, explicitly state that the context does not provide enough information, rather than guessing.

Answer in one or two complete sentences.

## Module 6 · Structured Output & Prompt Formatting

*Bullet Points, Tables, JSON, XML, Schemas & Output Validation*

### Q33. Prompt

Explain the advantages of remote work in exactly 6 bullet points.

Each bullet point should focus on a distinct advantage (for example, flexibility, cost savings, productivity, work-life balance, access to wider talent, and reduced commute stress) without overlapping content between points.

Do not include any introductory sentence, concluding paragraph, or additional explanation before or after the list.

Keep each bullet point to a single concise sentence of no more than 15-20 words.

Return only the 6 bullet points and nothing else.

### Q34. Prompt

Compare 'Machine Learning' and 'Deep Learning' using a table format.

Include exactly three columns: Definition, Data Requirement, and Typical Use Case.

For each concept, keep the content in every cell to one or two short sentences so the table remains easy to scan at a glance.

Ensure the comparison highlights the key practical differences, such as Deep Learning typically requiring much larger datasets and more computing power than traditional Machine Learning.

Return only the table, with no additional explanatory text before or after it.

### Q35. JSON-Based Extraction Prompt

Extract structured candidate information from the resume paragraph below and return it strictly as JSON matching this schema: { "name": string, "role": string, "experience\_years": number, "skills": array of strings }.

Resume paragraph: "Rahul Mehta is a Backend Developer with 4 years of professional experience, skilled in Java, Spring Boot, MySQL, and REST API design."

Ensure the JSON is valid, uses double quotes for all keys and string values, and does not include any extra fields beyond the schema.

Do not include any explanatory text, comments, or markdown code fences — return only the raw JSON object.

If any field cannot be confidently determined from the text, use null for that field instead of guessing.

### Q36. XML-Based Extraction Prompt

Extract the same candidate information used previously and return it strictly as XML instead of JSON.



Use the following structure with appropriate opening and closing tags:

<candidate><name></name><role></role><experience\_years></experience\_years><skills><skill></skill></skills></candidate>.

Resume paragraph: "Rahul Mehta is a Backend Developer with 4 years of professional experience, skilled in Java, Spring Boot, MySQL, and REST API design."

Include one <skill> tag inside <skills> for each individual skill mentioned, rather than combining them into a single tag.

Return only the well-formed XML output, with no additional commentary, markdown formatting, or code fences.

### Q37. Schema-Based Extraction Prompt

Extract product review data from the text below and return it as a JSON object matching this schema: { "product\_name": string, "rating": number, "sentiment": string, "main\_complaint": string }.

Review text: "I bought the SmartFit Watch last week and honestly the battery life is disappointing — it barely lasts a day."

For any field that cannot be clearly and confidently determined from the review text (for example, if no numeric rating is given), return null for that field rather than guessing or inferring a value.

Do not fabricate information that is not explicitly present in the text, even if it seems plausible.

Return only the JSON object with no additional explanation, comments, or formatting outside the object itself.

### Q38. Multi-Constraint Structured Prompt

Analyze the following customer complaint and process it through three steps: classify it into exactly one of three fixed categories (Billing, Product, Delivery), identify the main issue in one short sentence, and suggest one practical solution.

Complaint: "I was charged twice for my last order and the item still hasn't shipped after two weeks."

Return the entire result as a single JSON object with exactly these fields: { "category": string, "main\_issue": string, "suggested\_solution": string }.

If the complaint touches on more than one category, choose the single most dominant issue for the 'category' field.

Return only the JSON object, with no additional text, explanation, or markdown code fences before or after it.

### Q39. CSV-Style Output Prompt

Convert the following business requirement into a CSV-style output listing sample matching records: "Find all customers who spent more than ₹10,000 last month."

Generate 5 realistic sample customer records that would satisfy this condition, since no actual database is connected.

Use clearly labeled columns: Customer\_ID, Customer\_Name, Total\_Spend\_Last\_Month, and Region.

Format the output as comma-separated values with the header row first, followed by one row per sample record, and ensure every listed 'Total\_Spend\_Last\_Month' value is above ₹10,000 to match the requirement.

Return only the CSV-formatted text, with no additional explanation before or after it.

## Module 7 · Prompt Engineering for Coding

*Code Generation, Explanation, Debugging, Optimization, Documentation & Testing*

### Q40. Code Generation Prompt

Role: Act as a senior Python developer who writes clean, well-documented, production-quality code.

Language: Python 3. Task: Write a function called is\_prime(number) that checks whether a given number is a prime number.

Requirements: The function must correctly handle edge cases such as 0, 1, and negative numbers by returning False (since primes are defined only for positive integers greater than 1), and must raise a clear error or return a message if a non-integer input is provided.

Input: A single integer (or a non-integer, for error handling). Output: A boolean value (True or False), or an appropriate error/message for invalid input.

Include a short docstring explaining the function's purpose, parameters, and return value, and add 2-3 example calls with expected outputs as comments below the function.

### Q41. Code Explanation Prompt

Act as a patient programming tutor explaining code to a complete beginner who has never written a line of code before.

Explain the following Python code line by line: [insert the chosen 5+ line piece of Python code here, for example a simple loop that calculates the sum of a list of numbers].

For each line, explain both what the line does technically and why it is written that way, avoiding assumptions about prior programming knowledge.

Use simple analogies where helpful (for example, comparing a loop to repeating a task multiple times) to make abstract concepts concrete.

Present the explanation as a numbered list matching each line of code, followed by a short one-paragraph summary of what the overall code accomplishes.

### Q42. Debugging Prompt

Act as an experienced software engineer helping debug a piece of Python code.



Buggy code: [insert the short buggy code snippet here, for example a function that divides two numbers without handling division by zero].

Exact error message produced: [insert the exact error message and traceback text here, for example 'ZeroDivisionError: division by zero'].

Expected output: [describe what the code should correctly return instead, for example 'the function should return None and print a friendly warning when dividing by zero'].

Explain clearly why this error occurs, then provide the corrected code with the fix applied, followed by a one-line summary of what was changed and why.

### Q43. Code Optimization Prompt

Act as a senior Python developer focused on writing efficient, high-performance code.

Improve the performance of the following Python function while preserving its exact output and behavior for all valid inputs: [insert the given Python function here, for example an inefficient nested-loop function that finds duplicate values in a list].

Do not change the function's name, parameters, or the format of its return value — only optimize the internal logic (for example, replacing nested loops with a more efficient data structure such as a set or dictionary).

Explain the changes made using a clear before-and-after comparison, describing why the new approach is more efficient (for example, in terms of time complexity).

Present the response as: 'Original Code', 'Optimized Code', and 'Explanation of Changes', in that order.

### Q44. Code Refactoring Prompt

Act as a senior developer performing a code-quality review and refactor.

Refactor the following messy piece of code by adding proper functions where logic is repeated or unstructured, renaming variables to be meaningful and descriptive, and formatting the entire code according to PEP 8 style guidelines: [insert the messy code snippet here].

Do not change the code's final output or behavior in any way — the refactored version must produce identical results to the original for all valid inputs.

Add brief docstrings or comments only where they genuinely improve readability, without over-commenting obvious lines.

Present the response as 'Original Code' followed by 'Refactored Code', so the improvements can be compared side by side.

### Q45. Unit Test Generation Prompt

Act as a QA engineer writing thorough unit tests using Python's unittest framework.

Write unit tests for a function called `calculate_average(numbers)` that calculates the average of a list of numbers.

Include exactly 3 normal test cases (typical lists of positive numbers), 2 edge cases (such as a single-element list and a list containing negative numbers or zero), and 2 invalid-input test cases (such as an empty list and a non-list input like a string), asserting the expected exception or return value for each.

Use clear, descriptive test method names that indicate what each test is verifying (for example, `test_average_of_positive_numbers`).

Structure the output as a complete, runnable `unittest.TestCase` class, including the necessary import statement and an `if __name__ == '__main__': unittest.main()` block at the end.

### Q46. SQL Generation Prompt

Act as a database engineer who writes accurate, efficient SQL queries.

Convert the following business requirement into a correct SQL query: "List all employees whose department is Sales and salary is above ₹60,000."

Table schema: a table named `employees` with columns `employee_id`, `name`, `department`, and `salary`.

Select only the `employee_id`, `name`, and `salary` columns in the result, and order the results by salary in descending order.

Return only the SQL query as clean, properly formatted SQL, with no additional explanation unless explicitly requested.

## Module 8 · Prompt Engineering for Data Analysis

*Data Understanding, Cleaning, EDA, Visualization, ML Model Selection & Business Insights*

### Q47. Dataset Understanding Prompt

Act as a senior data analyst reviewing a new dataset before beginning any analysis.

Dataset description: A retail sales dataset with approximately 50,000 rows and 8 columns, including `Order_Date`, `Product_Category`, `Region`, `Quantity`, `Unit_Price`, `Revenue`, `Customer_ID`, and `Payment_Method`, where some rows have missing values in `Region` and `Payment_Method`.

Based on this description, describe the likely number of rows and columns, the probable data type of each column (numeric, categorical, date, or text), and where missing values are likely to cause problems.

Identify at least three possible data quality issues to investigate, such as duplicate orders, inconsistent category naming, or outlier values in `Revenue`.

Present the response with clear headings: 'Structure Overview', 'Data Types', 'Missing Values', and 'Potential Data Quality Issues'.

### Q48. Data Cleaning Prompt

Act as a data analyst responsible for preparing a dataset for analysis.

Dataset description: The same retail sales dataset described above, where the Region column is missing for about 4% of rows and the Payment\_Method column is missing for about 7% of rows.

Identify which columns have missing values and, for each one, suggest the most appropriate treatment method (for example, imputing with the most frequent category, creating an 'Unknown' category, or removing rows if the missing percentage is very small).

Justify why the suggested method is appropriate for that specific column, considering the column's data type and its likely importance to later analysis.

Present the response as a short table with columns: Column Name, Missing %, Suggested Treatment, and Justification.

#### **Q49. EDA Prompt**

Act as a data analyst performing exploratory data analysis on a sales dataset containing the columns Date, Product, Region, and Revenue.

Identify important patterns in the data, such as seasonal trends in Revenue over time, top-performing Products, and Regions with consistently high or low sales.

Identify potential outliers, such as unusually high or low Revenue values on specific dates, and suggest possible explanations for them (for example, promotional events or data entry errors).

Describe likely relationships between variables, such as whether certain Products perform differently across different Regions or time periods.

Present the findings under three headings: 'Key Patterns', 'Potential Outliers', and 'Variable Relationships', using concise bullet points under each.

#### **Q50. Data Visualization Prompt**

Act as a data visualization specialist recommending charts for the sales dataset containing Date, Product, Region, and Revenue columns.

Suggest three appropriate charts, one at a time, each addressing a distinct business question.

For each chart, specify: the chart type (for example, line chart, bar chart, or heatmap), what should be on the X-axis, what should be on the Y-axis, and the specific business question the chart helps answer (for example, 'Which region generates the most revenue?').

Ensure the three charts collectively cover trends over time, comparison across categories, and a relationship between two variables, rather than three similar chart types.

Present the response as a numbered list, one entry per chart, with each of the four details (type, X-axis, Y-axis, business question) clearly labeled.

#### **Q51. ML Model Selection Prompt**

Act as a machine learning consultant advising a business team on model selection.

Dataset description: A customer dataset with features such as tenure, monthly charges, contract type, and support tickets, with a binary target variable called Customer\_Churn (Yes/No).

Suggest two to three suitable machine learning algorithms for this classification problem (for example, Logistic Regression, Random Forest, and Gradient Boosting), briefly explaining why each is appropriate given the dataset's size and mix of categorical and numeric features.

Describe key preprocessing steps required before modeling, such as encoding categorical variables, scaling numeric features, and handling class imbalance if churn cases are rare.

Recommend appropriate evaluation metrics for this problem (such as precision, recall, F1-score, and ROC-AUC) and briefly explain why accuracy alone would be insufficient for an imbalanced churn dataset.

#### **Q52. Business Insight Prompt**

Act as a senior business analyst presenting findings to company leadership, not just a data reporter listing numbers.

Using the retail sales dataset described earlier (Date, Product\_Category, Region, Revenue, and related fields), analyze the data from a strategic business perspective rather than simply describing statistics.

Identify important trends (such as growing or declining product categories), risks (such as over-reliance on a single region or customer segment), and opportunities (such as underperforming regions with growth potential).

For each insight identified, include one specific, actionable recommendation that leadership could act on within the next quarter.

Present the response under four headings: 'Trends', 'Risks', 'Opportunities', and 'Recommended Actions', using concise bullet points suitable for an executive summary.

#### **Q53. Reusable Data Analysis Prompt Template**

Act as a [ROLE — e.g., senior data analyst / data scientist / BI consultant] analyzing a new dataset for our team.

Dataset description: [DATASET DESCRIPTION — briefly describe the dataset's source, approximate size, and key columns].

Business objective: [BUSINESS OBJECTIVE — state the specific business question or goal this analysis should support, such as improving retention or optimizing pricing].

Tasks: [TASKS — list the specific analysis tasks required, such as data quality review, exploratory analysis, visualization recommendations, model suggestions, or business insights].

Output Format: Present findings under clearly labeled headings matching the tasks requested, using concise bullet points or tables, and end with a short 'Key Takeaways' section summarizing the most important 2-3 points for a non-technical stakeholder.